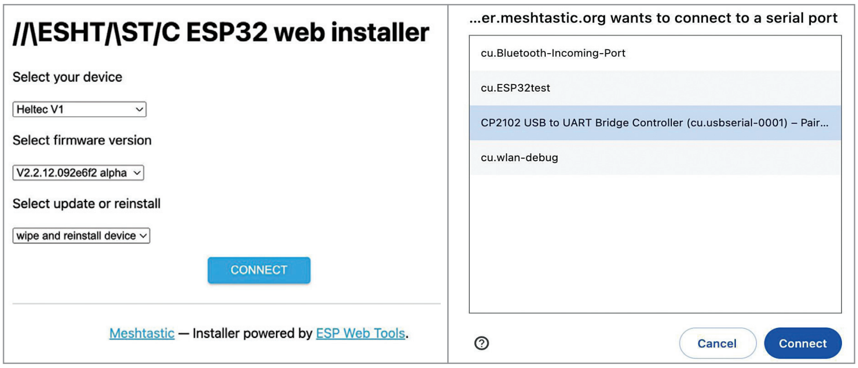

Fig. 4: Flashing firmware

```
VERSION.bin (for installing)
device-update.bat -f firmware-BOARD-
VERSION.bin(for updating)
```

Testing

To set up and test the project, you need to install Meshtastic app in your Android/iPhone (or tablet). There are Android and iOS versions available in their respective software stores (Google's Play Store/ Apple's App Store). To work with the three nodes, fix one mobile phone (or tablet) to play with each node. That way many setup jolts would be avoidable. The app is free! Fig. 5 shows the Meshtastic icon.

First, power on the Heltec board. Next, switch on Bluetooth in your mobile phone and pair with the Heltec board. The 6-digit pairing number will appear on the OLED screen. After successful pairing, open the Meshtastic app and you will find the app is connected [look for the tick mark on top right] with the board's Mac address. You can set a name for the board in the radio-configuration by pressing on the three dots (…) on right top of the app, as shown in Fig. 6.

The homemade Heltec is up now. Name it 'fabr' (short for fabricated one). The other two are not up. That is why question marks appear against their names. The tick mark on the top right cloud indicates that this node is connected.

Attaching sensors

Right now, only I2C sensors (on addresses 0x76, 0x77, 0x78, 0x18, 0x40, 0x41, 0x5D, 0x5c, 0x70, 0x44, 0x12, 0x3c are fixed for OLED) are allowed. More sensors will be added in the coming days. The Radio-configuration→Detection Sensor→Detection Sensor should be enabled.

You may provide a name to the sensor (BME or Temp_Ind). In the same window you can also designate a GPIO pin to monitor for [high or low] and the detection will be monitored periodically every 900 seconds (default value). Please note this GPIO pin status is totally independent of the sensor value. Fig. 7 shows the APP UI showing environment matrix log.

Fig. 5: Meshtastic icon

In Radio Configuration→Device →Role (for transmitting sensor value), name it 'SENSOR'. The other options are 'CLIENT' or 'ROUTER_CLIENT,' 'REPEATER' etc. Fig. 8 shows the APP UI showing devices in Mesh.

To send signals to remote hardware, you must define the GPIO pin first. However, only digital read and digital write is enabled here.

External notifications

For external notifications, go to Radio Configuration→External Notification→Enable. You also must set the output GPIO pins which will receive the notifications. You can set LED/Alarm Buzzer/Alarm Vibrator on three different GPIO pins along with time duration of the alarm.

In the demo 'fab' unit the alarm on GPIO-16 is set to light an LED for two seconds. The sensor values will appear on the Android/iOS screen and on the OLED screen. To add a new node, just set the node up and it will be added in the network very easily.

GPS location

If the board has GPS locator, it will deliver it automatically. Otherwise, if the Android/iOS device has location information turned on, it will deliver the GPS location, which will

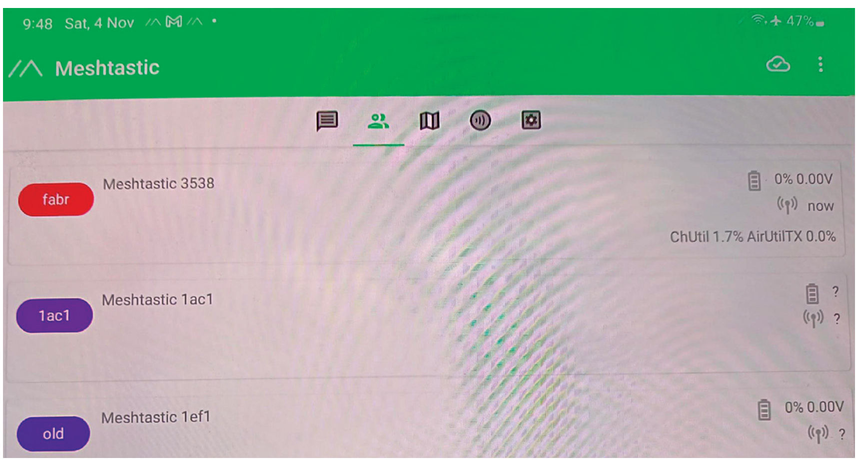

Fig. 6: Meshtastic app